



Research on efficient removal of ciprofloxacin through sequential rice straw biochar modification via alkali activation and manganese oxides

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ABSTRACT

Biochar is considered an environmentally friendly and cost-effective option for removing antibiotics from aqueous solutions. In this study, biochar derived from rice straw (RBC) was sequentially modified by KMnO_4 and NaOH (Mn/Na-RBC) to achieve ciprofloxacin (CIP) removal. To explore the adsorption mechanisms between Mn/Na-RBC and CIP, the characterization analyses, adsorption isotherms, thermodynamics, and kinetics were investigated. The results revealed that Mn/Na-RBC significantly increased the specific surface area ($154.12 \text{ m}^2 \cdot \text{g}^{-1}$), carbonization, and the number of oxygen-containing functional groups compared with pristine biochar, thereby substantially enhancing the CIP adsorption capacity. Through fitting of the adsorption model, it was found that the highest adsorption capacity was $32.25 \text{ mg} \cdot \text{g}^{-1}$ under acidic conditions ($\text{pH}=3$), and the pseudo-second-order and Freundlich model fit the adsorption behavior well. The main mechanisms of the adsorption process were pore filling, electrostatic interaction, H-bonding, π - π conjugation, n - π interaction and surface complexation. Moreover, the Mn/Na-RBC still exhibited a remarkable capacity to selectively adsorb CIP even in the presence of high ionic strengths. Thus, Mn/Na-RBC is a promising adsorbent for removing emerging contaminants from aquatic environments.

1. Introduction

Antibiotics are widely used in both human therapy and livestock farming and are essential antimicrobial agents (Danner et al., 2019). According to statistics, China has emerged as the world's largest consumer of antibiotics, with annual usage exceeding 25,000 tons (Zhang et al., 2020). The dispersion of antibiotics in the environment due to antibiotic abuse and improper disposal can have significant consequences for both human health and ecosystems (Danner et al., 2019; Li et al., 2023). Antibiotics are frequently detected in many water bodies, including groundwater, surface water and wastewater (Xu et al., 2021). Ciprofloxacin (CIP) is a third-generation fluoroquinolone compound that is frequently prescribed for the treatment of respiratory and urinary tract infections (Mathur et al., 2021). The incomplete metabolism of humans and animals, as well as the ineffective removal performance of traditional technologies such as wastewater treatment plants (WWTPs), results in the continuous discharge of antibiotics into natural water and soil through runoff and osmosis (Igwegbe et al., 2021; Wang et al., 2023; Zeng et al., 2019). According to reports, the CIP concentration

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of natural water bodies ranges from $0.5 \mu\text{g}\cdot\text{L}^{-1}$ to $6.5 \text{ mg}\cdot\text{L}^{-1}$ (Danner et al., 2019). However, it should be noted that concentrations as high as $30 \text{ mg}\cdot\text{L}^{-1}$ have been found in the effluents of wastewater treatment facilities for the pharmaceutical industry (Van Doorslaer et al., 2014). The continuous presence and release of antibiotics in aquatic systems can pose a threat to human health. This is because it can introduce multiple antibiotic resistance genes in bacteria, thereby increasing the risk of antibiotic resistance (Li et al., 2023). Hence, there is an urgent need to discover an economically feasible approach for eliminating emerging environmental pollutants from water sources.

Several technologies, including membrane technology (Qalyoubi et al., 2022), Fenton (Qian et al., 2020), photocatalytic degradation (Zong et al., 2021), biodegradation (Bueno et al., 2023), and adsorption (Theamwong et al., 2021), have been developed to remove antibiotics from water. Among them, adsorption is widely regarded as the most favorable option due to its high efficiency, low cost, ease of operation, and environmentally friendly nature. Several adsorbents, such as activated carbon (Sousa et al., 2021), graphene oxide (Ashraf et al., 2022) and metal–organic frameworks (MOFs) (Zhang et al., 2022), have been utilized for the removal of CIP from water. Nonetheless, their utilization has been restricted due to the intricate preparation procedure and the associated high expenses (Xu et al., 2022). In comparison, biochar is a promising adsorption material because of its advantages of a well-developed porous structure and large surface area (Biswal and Balasubramanian, 2022; Xiang et al., 2019).

However, regarding the adsorption of emerging contaminants, pristine and unmodified biochars are still limited by their functional groups and porosity (Du et al., 2023). Thus, pristine biochar can be optimized in terms of its physicochemical properties through modifications, including metal, acid, and base modifications, non-metallic atomic doping, aging, and ball-milling (Anuar et al., 2023; Qiu et al., 2022). In contrast to other physical and chemical modification methods, metal modification is utilized to enhance the surface area and functional groups of biochar to improve its adsorption capacity. According to recent studies, KMnO_4 is an effective activator for controlling the pore structure and elemental composition of biochar due to its strong oxidizing properties (He et al., 2021; Zheng et al., 2021). Moreover, the use of biochar as a substrate can effectively inhibit the aggregation of manganese oxides (MnO_x) (Shaheen et al., 2022). In addition, MnO_x , which contains active hydroxyl groups, can effectively function as an adsorbent for removing pollutants from water (Tan et al., 2016). However, the presence of MnO_x in biochar may affect its pore structure and surface area (Shaheen et al., 2022). Faheem et al. (2016) found that the modification of biochar using MnO_x increased the specific surface area from 234.8 to $340 \text{ m}^2\cdot\text{g}^{-1}$, and decreased the pore width from 0.101 to 0.056 nm . Hence, the inclusion of an additional activator is essential for optimizing the pore structure of biochar. Alkali modification can dissolve the residual melt inside the pores, resulting in substantial increases in the specific surface area and porosity of biochar (Du et al., 2023), and thus it may be an ideal activator for pore development of biochar.

In this study, we utilized KMnO_4 modification of rice straw biochar (Mn-RBC) to enhance the adsorption capacity of biochar by creating more adsorption sites. Subsequently, NaOH modification was employed to increase the surface area of the biochar and generate additional microporous structures (Mn/Na-RBC). Rice straw biochar modification and its application in CIP adsorptive removal have not previously utilized the aforementioned methods. Therefore, the objectives of this study were i) to examine the impact of single (KMnO_4) and dual ($\text{KMnO}_4/\text{NaOH}$) chemical activation treatments on the characteristics of biochars; ii) to investigate the adsorption behavior of CIP onto Mn/Na-RBC, through a series of experiments on the including isotherms, kinetics, thermodynamics and environmental factors experiments; and iii) to reveal the possible adsorption mechanisms of CIP on Mn/Na-RBC through models and characterization analysis.

2. Materials and method

2.1. Chemicals

The CIP ($\text{C}_{17}\text{H}_{18}\text{FN}_3\text{O}_3$, $\geq 98\%$) was purchased from Aladdin (Shanghai, China) and used to prepare the stock solution. Potassium permanganate (KMnO_4 , Aladdin, $>99\%$) and sodium hydroxide (NaOH , Aladdin, 98%) were used to modify the prepared biochar. All of the solutions were prepared using ultra-pure water.

2.2. Preparation of RBC, Mn-RBC, and Mn/Na-RBC

The detailed preparation processes of the RBC, Mn-RBC and Mn/Na-RBC are follows. (i) RBC: the ground and dried rice straw was placed into tube furnace and pyrolyzed under an N_2 -atmosphere at 500°C for 2 h with a heating rate of $10^\circ\text{C min}^{-1}$ to obtain the pristine rice straw biochar (RBC). (ii) Mn-RBC: for KMnO_4 modification alone, 5 g of RBC was mixed with KMnO_4 solution (100 mL, 0.1 M) via magnetic stirring at 500 rpm for 24 h. The reaction mixture was filtered and dried at 80°C until a constant weight was achieved (RBC/ KMnO_4). The obtained sample was heated at 700°C for 1 h (denoted as Mn-RBC). (iii) Mn/Na-RBC: the RBC/ KMnO_4 mixture was immersed in 100 mL of deionized water containing 1 M NaOH and stirred continuously for 12 h. The solution was then filtered and dried at 80°C . The dried sample was subject to high-temperature pyrolysis according to the procedures in (ii) for biochar preparation. The resulting product is denoted as Mn/Na-RBC. Following this, the products (i.e., RBC and Mn-RBC) were subjected to a rinsing process using 0.1 M HCl to eliminate any remaining traces of NaOH and impurities. Subsequently, they were rinsed multiple times with deionized (DI) water until a neutral pH was achieved, after which the solids were dried until a consistent weight was attained.

Table 1
Element contents and ratios of biochars.

Biochars	Elements (%)				H/C	O/C	(N+O)/C
	C	H	O	N			
RBC	42.9	1.77	6.44	1.43	0.0418	0.1519	0.1857
Mn-RBC	40.17	0.93	6.22	1.24	0.0232	0.1548	0.1857
Mn/Na-RBC	27.38	0.48	5.46	0.77	0.0175	0.1994	0.2275

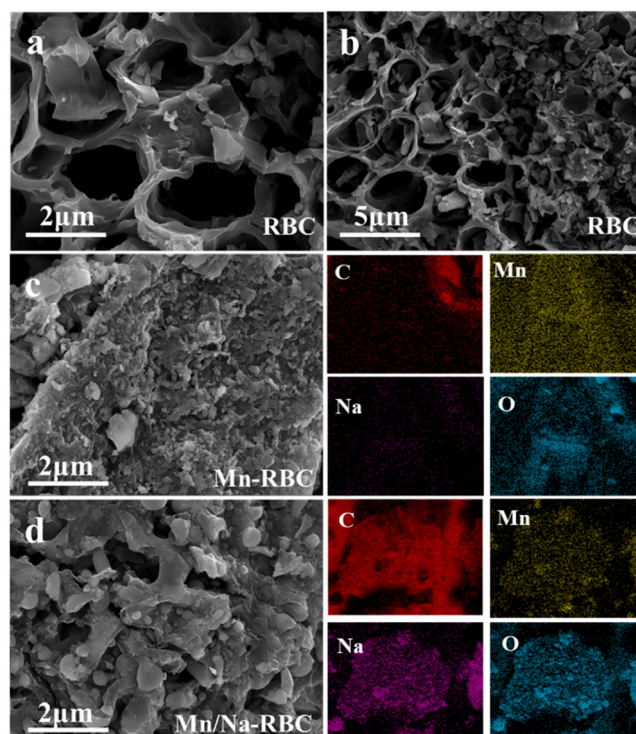


Fig. 1. SEM images of RBC (a-b); SEM-EDS images of Mn-RBC (c), Mn/Na-RBC (d).

2.3. Characterization

The surface morphologies and elemental composition of the biochars (i.e., RBC, Mn-RBC, and Mn/Na-RBC) were analyzed via scanning electron microscopy (SEM) (LMS, TESCAN MIRA, Czech Republic) and energy dispersive spectrometry (EDS). The Surface area and pore volume were measured using Micromeritics ASAP 2460 (USA) based on the Brunauer–Emmett–Teller (BET) method. The crystallographic structures of the biochars were determined via X-ray diffraction analysis (XRD) (SmartLab SE, Rigaku, Japan). X-ray photoelectron spectroscopy (XPS) (K-Alpha, Thermo Scientific, USA) was employed to investigate the surface compositions of the biochars. Fourier transform infrared (FT-IR) (Nicolet iS5, Thermo Scientific, USA) was used to determine the surface organic functional groups, and the measured wavenumber range from 400 to 4000 cm^{-1} . The element contents of the biochars were determined using an elemental analyzer (Flash SMART, Thermo Scientific, USA). The zero-point charges (pH_{zpc}) of the biochars were determined following the method of [Zhu et al. \(2014\)](#).

2.4. Adsorption experiments

The adsorption of CIP onto RBC, Mn-RBC and Mn/Na-RBC was studied through batch adsorption. Adsorption experiments were conducted using 50 mL plastic centrifuge tubes. The tubes were placed in a thermostatic oscillator and shaken at a speed of 220 rpm under dark conditions. The initial pH was adjusted between 3.0 and 10.0 using 0.1 M HCl or NaOH as necessary. The collected samples after adsorption equilibrium were filtered with 0.45 μm syringe filter and their residual concentrations were measured by UV–vis spectrophotometer at $\lambda = 270 \text{ nm}$. All experiments were carried out in triplicate. The specific reaction parameters of the adsorption experiment are as follows:

Effect of pH and ionic strength: C (CIP) = 10 $\text{mg}\cdot\text{L}^{-1}$, V=20 mL, m = 15 mg, t= 12 h, T = 298 K, C (Na^+ , K^+ , Ca^{2+} , Cd^{2+}) = 0, 0.01, 0.05 and 0.01 $\text{mol}\cdot\text{L}^{-1}$.

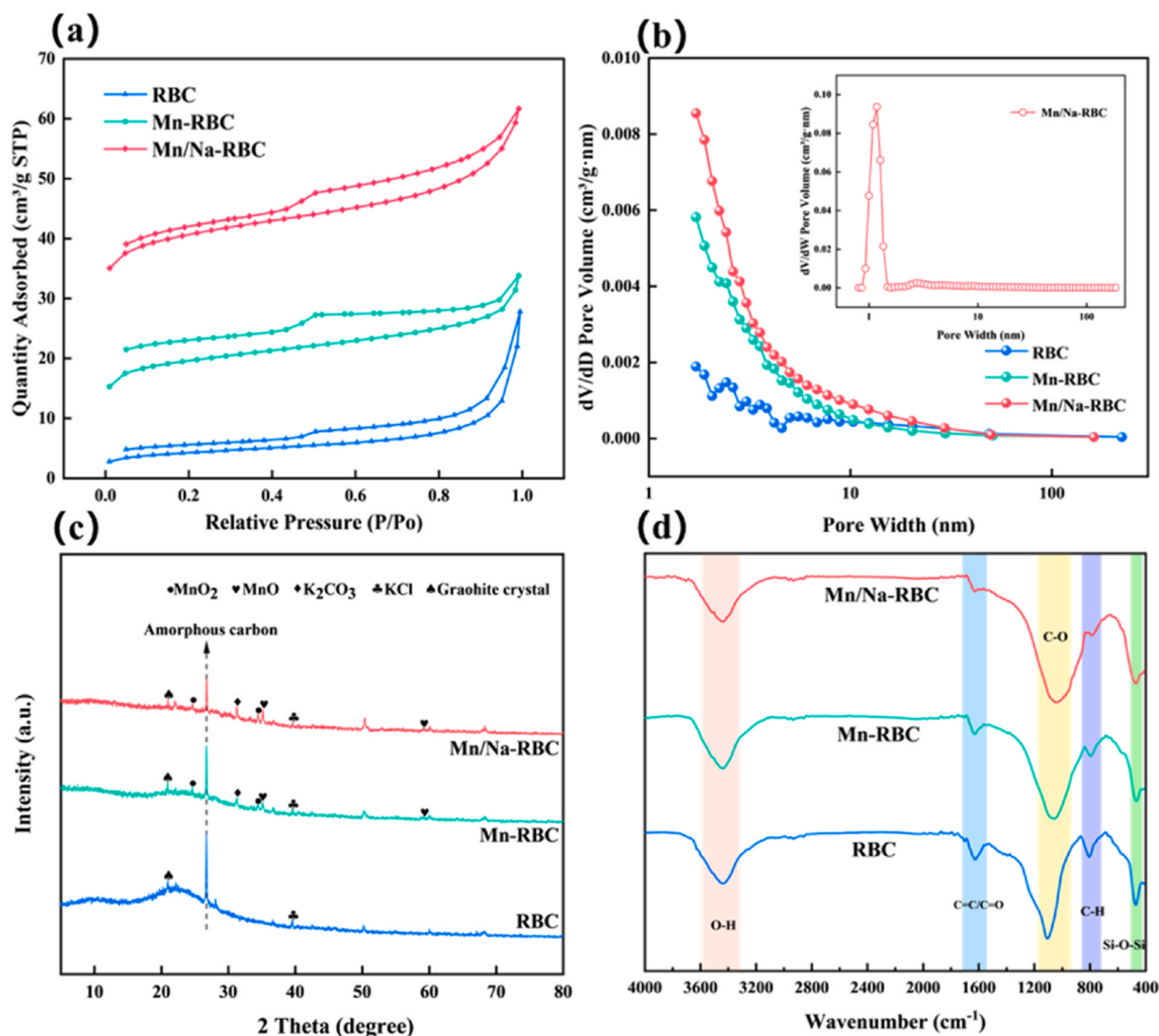


Fig. 2. The N_2 adsorption-desorption curves (a), pore size distribution (b), XRD patterns (c), FT-IR spectrum (d).

Adsorption isotherms and thermodynamics: C (CIP) = 5–200 $\text{mg}\cdot\text{L}^{-1}$, $V=20$ mL, $m=15$ mg, $t=12$ h, $T=298$ – 318 K.

Adsorption kinetics: C (CIP) = 10 $\text{mg}\cdot\text{L}^{-1}$, $V=20$ mL, $m=15$ mg, $t=0$ – 24 h, $T=298$ K.

2.5. Model fitting

The relevant formulas, adsorption kinetics, adsorption isotherms models and their fitting data were showed in [Supplementary data](#).

3. Results and discussion

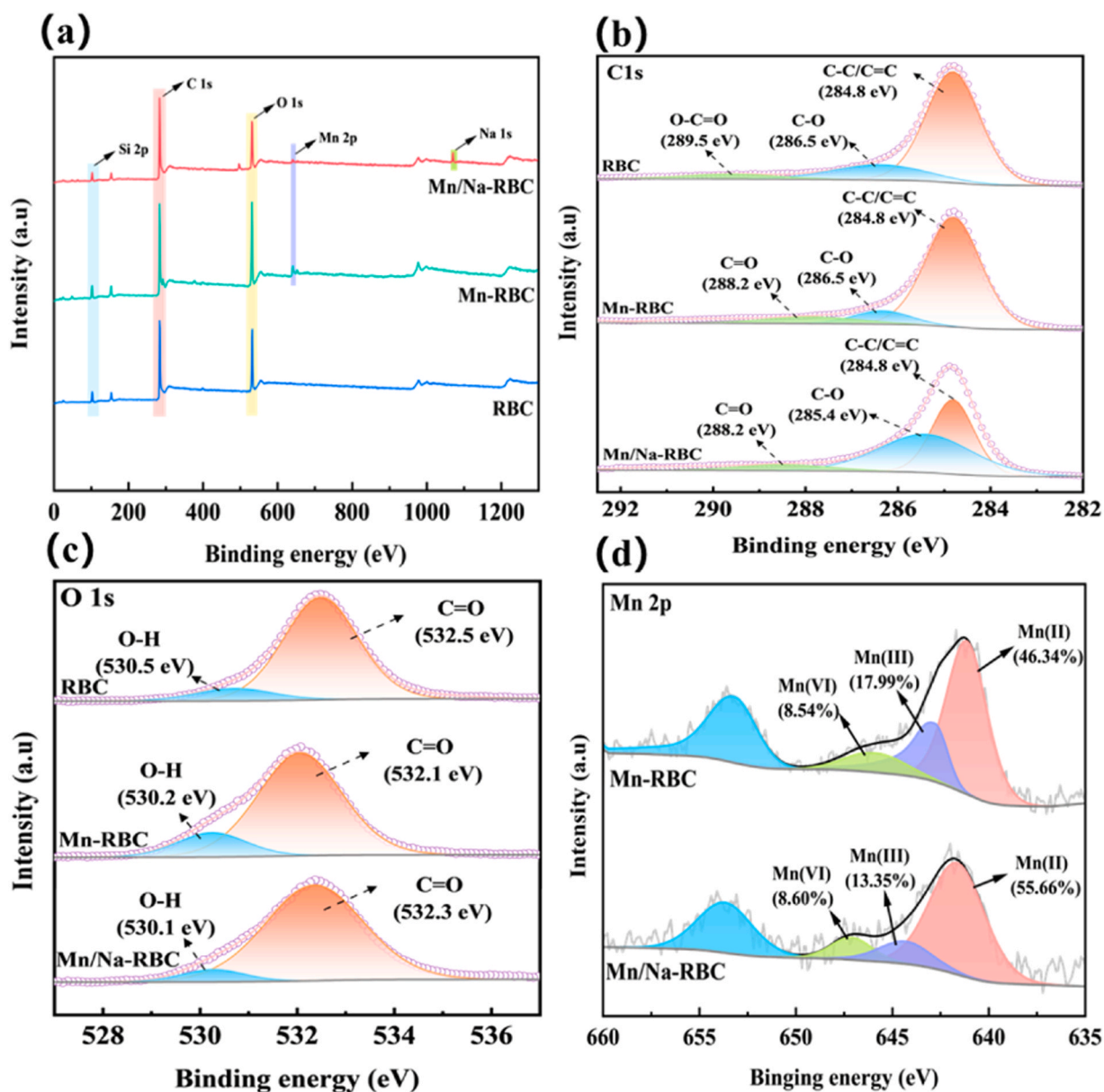
3.1. Characterization of biochars

The element compositions of the biochars are presented in [Table 1](#). The C, H, and O contents of the Mn-RBC and Mn/Na-RBC decreased compared with those of the pristine RBC, which may have been due to the strong oxidizing effect of KMnO_4 and the chemical reaction between the RBC, KMnO_4 , and NaOH (Xu et al., 2022). The changes in the electronegativity could either enhance or impede the electrostatic interactions between the adsorbent and adsorbate. The O/C, H/C, and (N+O)/C atomic ratios can be used as indicators of the hydrophilicity, aromaticity, and polarity of biochars, respectively (Rajapaksha et al., 2014). The H/C ratio decreased due to the demethylation reaction via the loss of $-\text{CH}_3$ (Biswas et al., 2022). The lowest H/C ratio was observed in the Mn/Na-RBC, indicating a higher degree of carbonization and increased aromaticity of the biochar (Biswas et al., 2022). The increases in the O/C and

Table 2

Surface area and porosity of RBC, Mn-RBC and Mn/Na-RBC.

Biochar	Specific surface area ($\text{m}^2\cdot\text{g}^{-1}$)	Total pore volume ($\text{cm}^3\cdot\text{g}^{-1}$)	Average pore width (nm)
RBC	14.85	0.0403	11.55
Mn-RBC	72.36	0.0298	2.40
Mn/Na-RBC	154.12	0.0436	2.20

**Fig. 3.** XPS survey (a) and spectra of C 1 s (b), O 1 s (c), Mn 2p (d).

(N+O)/C ratios following the modification, suggest enhancement of the hydrophilicity and polarity. Moreover, the increase in the O/C ratio indicates augmentation of the oxygen-containing functional groups, contributing to the negative charges of the Mn-RBC and Mn/Na-RBC (Biswas et al., 2022; Yin et al., 2023). The excellent aromaticity and polarity facilitated the adsorption of pollutants by the adsorbent.

The morphological characteristics and distributions of the main elements of the adsorbent materials were determined via SEM-EDS.

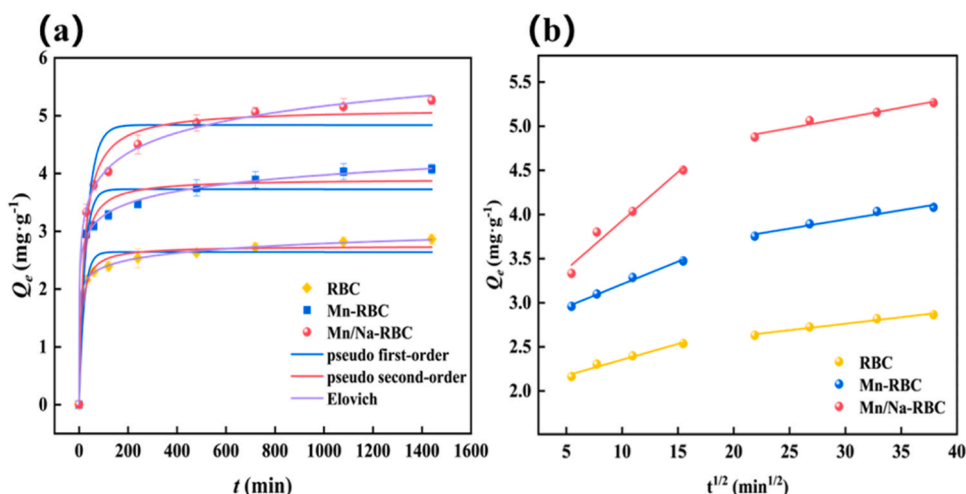


Fig. 4. Adsorption kinetics of CIP adsorbed onto biochars. ([BC]₀=15 mg, [CIP]= 10 mg.L⁻¹, [pH]= 3, and [T]= 298 K).

Images were showed in Fig. 1. The surface morphology of the RBC (Fig. 1a and b) was characterized by an irregular shape, with more abundant slit-shaped particles compared with the Mn-RBC and Mn/Na-RBC. The SEM images (Fig. 1c and d) show that the surfaces of the Mn-RBC and Mn/Na-RBC were rough, which was thought to be caused by the manganese oxide particles (MnO_x) (Yin et al., 2020). The C, O, Na, and Mn were observed to have uniform distributions via EDS, and the modified biochars had higher Mn and O contents compared with those of the RBC. The above results indicated that MnO_x was successfully loaded onto the surface of the RBC, resulting in a larger specific surface area and more active sites, which improved its CIP adsorption capacity.

The N₂ adsorption–desorption isotherm and pore size distribution were important parameters to judge the adsorption performance of adsorbents. The results are shown in Fig. 2a and b. The N₂ adsorption–desorption isotherms of all of the samples exhibited a combination of type I and type IV isotherms. The N₂ adsorption volume noticeably increased at extremely low P/P₀ values, reflecting the presence of a well-developed microporous structure. In addition, the capillary condensation took place at P/P₀ = 0.5–0.8 and formed a hysteresis loop, which suggested the presence of a mesoporous structure (Yang et al., 2016). Table 2 presents a summary of the surface area and pore volume parameters. Compared with the surface area of the RBC (14.85 m².g⁻¹), the surface areas of the Mn-RBC and Mn/Na-RBC were 72.36 and 154.14 m².g⁻¹ after modification with KMnO₄ and NaOH. The presence of manganese oxides and rinsing in an alkali solution were conducive to improve the specific surface areas of the biochars (Liu et al., 2021; Tang et al., 2018). The pore size decreased from 11 (RBC) to ~2 nm (Mn-RBC and Mn/Na-RBC), indicating that the strong oxidation led to the loss of organic matter (Yin et al., 2023). Notably, the Mn/Na-RBC biochar exhibited a highly microporous structure. The pore diameters of the RBC, Mn-RBC, and Mn/Na-RBC were all larger than the CIP (length × width × height: 1.35 × 0.30 × 0.74 nm) (Sun et al., 2016), suggesting that the influence of the pore size resistance could be disregarded (Ma, Y. et al., 2021). The superior specific surface area and pore volume of the Mn/Na-RBC provided channels for ion exchange and diffusion.

The crystallinity of the RBC, Mn-RBC, and Mn/Na-RBC was analyzed via XRD (Fig. 2c). The XRD patterns revealed that all of the samples possessed sharp diffraction peaks at approximately 26°, which corresponded to amorphous carbon (Yan et al., 2020). The rice straw contained high levels of potassium (K), which led to the production of KCl (2θ=39.4°) after pyrolysis (Qu et al., 2023). After modification, the new peak at 24.7°, 34.4°, 35.1°, 59.1°, and 31.2° appeared in the XRD patterns of the Mn-RBC and Mn/Na-RBC, suggesting the formation of MnO₂, MnO and K₂CO₃ (Fan et al., 2018; Zhang et al., 2023; Zhu et al., 2021). The characteristic peaks around 20° corresponded to the formation of a graphitic structure (Ma et al., 2021). The graphitic structure, serving as a π-electron donor, was capable of forming π-π conjugation with CIP, which acted as a π-electron acceptor (Ma et al., 2021).

The functional groups of the RBC, Mn-RBC, and Mn/Na-RBC were analyzed via FT-IR spectroscopy (Fig. 2d). It was found that the various adsorption peaks of the biochars included the vibration of C–O (1035–1101 cm⁻¹), C=C/C=O (1620–1624 cm⁻¹), and the symmetric stretching vibration of Si–O–Si (462–465 cm⁻¹), respectively (Wang et al., 2020; Wu et al., 2020; Zhu et al., 2014). The broad band at 3435 cm⁻¹ represented the stretching vibration of the hydrogen group (–OH) (Xiao et al., 2014). The peak observed at 789–801 cm⁻¹ was attributed to the flexural vibration of the C–H bonds within the benzene ring (Yang et al., 2020). The alcohol C–O stretching vibration was altered as a result of the KMnO₄ and NaOH modification. This modification resulted in the establishment of hydrogen bonding interactions between the –OH and C–O groups in the Mn/Na-RBC and the F, N, and O-containing groups in the CIP, thereby enhancing the CIP adsorption capacity (Ma et al., 2021).

XPS analysis was conducted to analyze the element compositions and the number and distribution of the functional groups on the adsorbent surface before and after modification (Fig. 3a–d). After deconvolution of the C 1 s peak, four chemical states were generated and assigned to O–C=O (289.5 eV), C=O (288.2 eV), C–O (286.5 eV), and C–C/C=C (284.8 eV), respectively (Chen et al., 2020). Compared with the RBC and Mn-RBC, the relative proportion of C–C/C=C decreased, and the proportion of C–O increased in the Mn/Na-RBC. This implied that the proportion of the C–C/C=C fraction was transformed into C–O due to the KMnO₄ and NaOH modification (Ma et al., 2021). Furthermore, the typical Mn 2p_{1/2} peak and Mn 2p_{3/2} doublet were observed in the Mn-RBC and

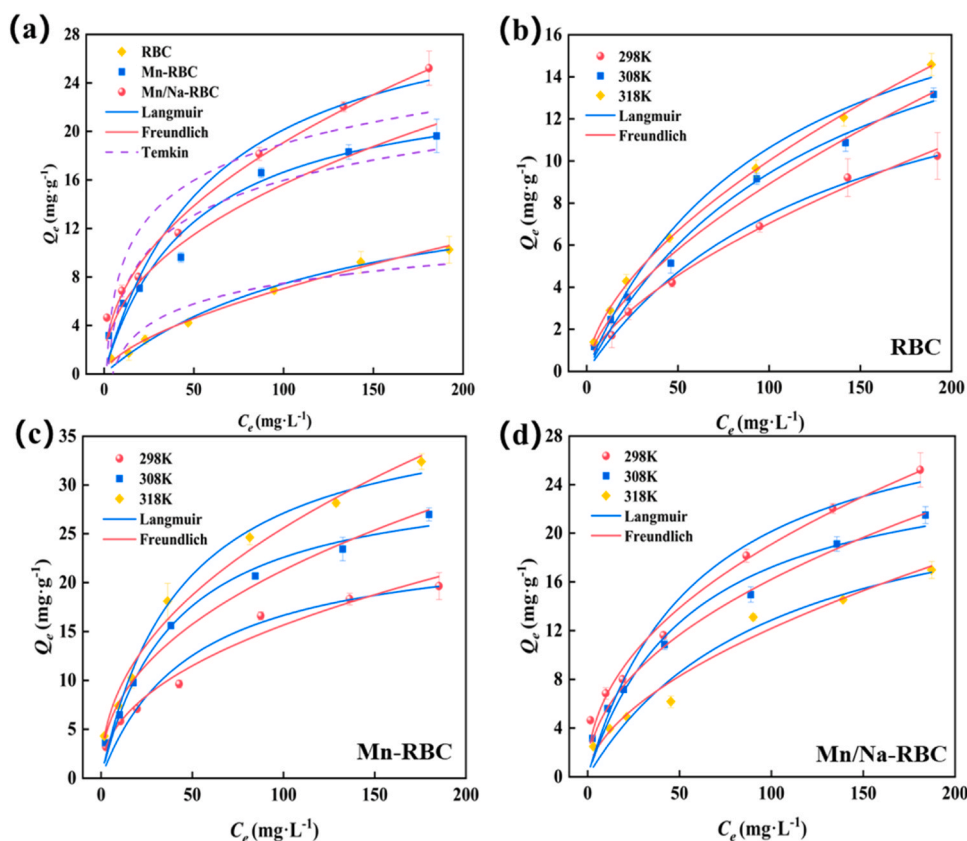


Fig. 5. Isotherms models (a) and thermodynamic models (b–d) fitting curves of CIP adsorption. onto RBC, Mn-RBC, and Mn/Na-RBC. ([BC]₀=15 mg, [CIP]= 10–200 mg·L⁻¹, [pH]= 3, and [T]=298, 308, 318 K).

Mn/Na-RBC. The Mn 2p_{3/2} peak was composed of three peaks: Mn (II) (641.9 eV), Mn (III) (644.6 eV), and Mn (IV) (647.8 eV) (Li et al., 2019). The oxidation states of Mn (III, IV) can induce lattice defects such as oxygen vacancies, resulting in the increased presence of hydroxyl oxygen (Zhang et al., 2023). The XPS analysis also revealed that Mn/Na-RBC contained abundant oxygen-containing functional groups and active sites, which had the potential to enhance its CIP adsorption capacity through various mechanisms, including H-bonding and π - π conjugation.

3.2. CIP adsorption behaviors of RBC, Mn-RBC, and Mn/Na-RBC

3.2.1. Kinetics

The adsorption kinetics describe the relationship between the diffusion process and the contact time of the CIP at the solid–liquid interface, which can be used to investigate help explore the reaction pathways and adsorption mechanisms (Azzam et al., 2022). In this study, the adsorption kinetics of CIP on the biochars were modeled and analyzed using the pseudo-first-order, pseudo-second-order, and Elovich kinetic models. The corresponding parameters are presented in Fig. 4a and Table S1. The pseudo-first-order model suggests that the adsorption process is mainly governed by the resistance to mass transfer within the particles. By contrast, the pseudo-second-order model suggests that the rate of adsorption is influenced by either a chemical reaction or the square of the adsorbed site vacancies on the surface of the adsorbent (Zhang et al., 2020). Similarly, the Elovich model assumes that the adsorbent surface active sites are a non-homogeneous diffusion process dominated by chemisorption (Lyubchik et al., 2016). With increasing contact time, the adsorption process of CIP exhibited a rapid initial increase and then gradual stabilization. In the initial stage of adsorption, there was a rapid increase in the adsorption rate owing to the high initial concentration of the adsorbate, numerous unoccupied active sites, and available functional groups. Then, the adsorption entered a slow phase until saturation equilibrium was reached as the adsorption sites were gradually occupied by CIP. Compared with the pseudo-first-order model (R^2 : 0.9383–0.9613), pseudo-second-order model (R^2 : 0.9766–0.9867) and Elovich model (R^2 : 0.9982–0.9997) more accurately described the adsorption kinetics. This observation suggested that the adsorption process was chemisorption and could be divided into two distinct dynamic steps. Firstly, the CIP molecules formed a monomolecular layer on the adsorbent surface, and then the adsorbents were controlled in the CIP adsorption through the occurrence of shared or exchanged electrons (Zeng et al., 2019).

The diffusion mechanism from the solid–liquid interface to the interior of the adsorbent particles played a crucial role in determining the total rate of CIP adsorption. Consequently, the intraparticle diffusion model was employed to delve deeper into the

Table 3

The CIP adsorption capacities on other adsorbents.

Adsorbent	pH	$q_{\max}$ (mg g ⁻¹)	Method of determination	Reference
Mn/Na-RBC	3.0	32.25	Langmuir	This study
CoFe-LDH-modified sludge biochar	-	19	Langmuir	(Zheng et al., 2022)
Manganese oxide-loaded magnetic biochar	3.0	8.37	Langmuir	(Li et al., 2018)
Dopamine functionalized cobalt ferrite NPs	7.0	16.53	Langmuir	(Malik et al., 2019)
Humic acid coated magnetic biochar	-	8.68	Langmuir	(Zhao et al., 2019)
Carbon nanofiber	5.0	10.36	Langmuir	(Li et al., 2016)

diffusion mechanism and to identify the rate-limiting steps within the adsorption process (Hu et al., 2019). Generally, the diffusion process of the pollutants on the adsorbent was as follows. (1) Initially, the high initial concentration led to the facilitation of the transport of the adsorbate from the bulk solution to the external surface of the adsorbent via liquid film diffusion (stage I). (2) Subsequently, the remaining adsorbate continued to permeate from the surface of the adsorbent into the internal voids through particle diffusion until equilibrium was reached (stage II) (Nguyen et al., 2022). The total adsorption rate was controlled by a combination of liquid film diffusion and intraparticle diffusion (Ali et al., 2016). Furthermore, the fitting plots of the intraparticle diffusion model (Fig. 4b) contained two linear segments, neither of which intersected with the origin. This discovery suggests that neither membrane diffusion nor intraparticle diffusion was the sole limiting factor affecting the overall adsorption process (Tang et al., 2018). Moreover, the C value indicated the magnitude of the boundary layer thickness, the larger C value, the greater boundary layer effect (Lyubchik et al., 2016). The higher C_1 values observed for the Mn-RBC and Mn/Na-RBC indicated a more significant contribution to the membrane diffusion compared with the RBC. This could be due to their greater specific surface areas (Huang et al., 2014).

3.2.2. Isotherms

The Langmuir, Freundlich, and Temkin isotherm models were used to further reveal the adsorption mechanisms of CIP onto the RBC, Mn-RBC, and Mn/Na-RBC. For a constant adsorbent dosage, the equilibrium adsorption capacities of the RBC, Mn-RBC, and Mn/Na-RBC increased progressively with rinsing in terms of the initial concentration of CIP, which aided in the assessment of their potential adsorption capacities (Ma et al., 2020). The Langmuir isotherm model is based on homogeneous monomolecular layer chemisorption, while the Freundlich adsorption model describes multimolecular layer adsorption on non-homogeneous surfaces (Gunay et al., 2007). The Temkin model considers the interaction between the adsorbent and adsorbate, and extremely high and low concentrations values of the adsorbate in the solution phase are neglected (Al-Ghouti and Da'ana, 2020; Wang and Guo, 2020). The isotherms and parameters of the adsorption of CIP onto the biochar are shown in Fig. 5a and Table S2. The maximum CIP adsorption capacities of the Mn-RBC and Mn/Na-RBC were 24.77 and 32.25 mg g⁻¹, respectively, which were much higher than those of the other modified biochars (Table 3). Furthermore, the Freundlich model ($R^2 = 0.973\text{--}0.992$) fitted the CIP adsorption process much better than the Langmuir model ($R^2 = 0.925\text{--}0.987$), demonstrating that the active sites on the adsorbent's surface had a non-homogeneous distribution and the CIP adsorption was dominated by multilayer adsorption (Al-Ghouti and Da'ana, 2020). For the Freundlich model, the values of $1/n$ determine whether adsorption was favorable (Liu et al., 2019). This means that the adsorption of CIP onto the biochar was more favorable after composite modification because all of the $1/n$ values were less than 1. Based on the fitting results for the Temkin model, it was found that there was a weak linear relationship between the adsorbent and adsorbate, indicating that chemisorption occurred, but it was not the only dominant mechanism in the adsorption of CIP (Hu et al., 2019). Consequently, the CIP adsorption by the RBC, Mn-RBC, and Mn/Na-RBC was simultaneously controlled by physisorption and chemisorption (Tang et al., 2018).

3.2.3. Thermodynamics

The thermodynamics of the CIP adsorption onto the RBC, Mn-RBC, and Mn/Na-RBC at different temperatures and the relevant parameters are shown in Fig. 5b–d and Table S3. All of the ΔG values were negative, indicating that the adsorption reactions were spontaneous (Atugoda et al., 2021). Based on the results of previous studies, the adsorption process was physical ($-20 \text{ kJ mol}^{-1} < \Delta G < 0$) and the adsorption process was chemisorption ($-400 \text{ kJ mol}^{-1} < \Delta G < -80 \text{ kJ mol}^{-1}$) (Lin et al., 2020). Hence, it was reaffirmed that physical adsorption was the dominant CIP adsorption process, but chemisorption also had an effect on the adsorption process. This has been confirmed in previous studies (Zhang et al., 2019). The ΔH value usually determines whether the adsorption process is endothermic (positive value) or exothermic (negative value) (Atugoda et al., 2021). As the temperature increased from 298 to 318 K, the negative ΔH value of the Mn/Na-RBC indicated that the adsorption process was exothermic. All of the absolute values of ΔH were in the range of $2\text{--}40 \text{ kJ mol}^{-1}$ (Lin et al., 2020), indicating that the Van der Waals force and hydrogen-bonding force drove the adsorption process. The positive ΔS values for CIP removal by the adsorbents revealed that the randomness and affinity increased during the adsorption process (Atugoda et al., 2021).

3.3. Effects of initial pH and ionic strength

The pH of the solution governs the surface charge of the adsorbents and influences the molecular form in which the CIP exists, and

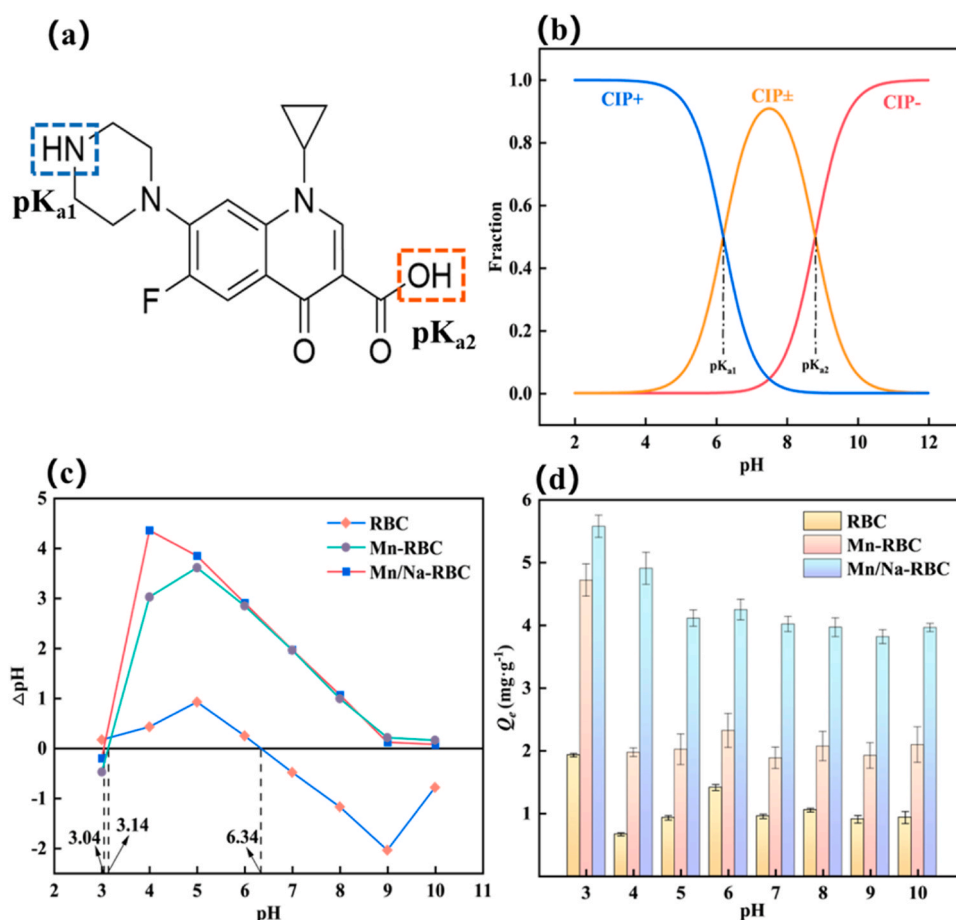


Fig. 6. Chemical structure of CIP (a); Species of CIP under different pH (b); zero-point charge of biochars at different pH (c); effect of pH on adsorption capacity of biochars for CIP (d). ([BC]₀=15 mg, [CIP]= 10 mg·L⁻¹, and [T]= 298 K).

thus it is regarded as a significant factor affecting the electrostatic interaction between adsorbents and adsorbate (Dai et al., 2019; Zhuang et al., 2015). The ionization of CIP under different pH values is shown in Fig. 6a and b, for two pK_a values of 6.2 and 8.8 and the main CIP species of CIP⁺ (pH < 6.2), CIP[±] (6.2 < pH < 8.8), and CIP⁻ (pH > 8.8) (Zheng et al., 2022). The effect of the initial pH of the solution on the CIP removal by the biochars is shown in Fig. 6d. As the pH increased from 3.0 to 10.0, the CIP adsorption capacities of the biochars gradually decreased. Significantly, the pHzpc values of the RBC, Mn-RBC and Mn/Na-RBC were 6.34, 3.04, and 3.14, respectively (Fig. 6c). When the pH was less than pHzpc, the surface charge of the RBC was protonated and positive, while the Mn-RBC and Mn/Na-RBC were negatively charged due to deprotonation (Tang et al., 2020). Under acidic conditions (pH=3), the adsorption of CIP by the Mn-RBC and Mn/Na-RBC was found to be favorable. Their CIP adsorption capacities were determined to be 4.72 and 5.58 mg·g⁻¹. This phenomenon was attributed to the tendency for acidic conditions to enhance the formation of hydrogen bonds, along with the presence of a slight negative charge on the surface of the adsorbents. This consequently enhanced the CIP adsorption efficiency through electrostatic attraction and H-bonding. Based on the above results, electrostatic interaction was one of the main mechanisms of the CIP adsorption.

The CIP adsorption capacities of the adsorbents were evaluated under varying cation concentrations (Na⁺, K⁺, Ca²⁺, and Cd²⁺). The results are shown in Fig. S1. The CIP binding abilities of the RBC and Mn-RBC were significantly enhanced as the Na⁺ and K⁺ concentrations increased. This was due to the enhancement of their hydrophobic interactions with the adsorbent, which was conducive to the adsorption of CIP (Zhang et al., 2010). The presence of Ca²⁺ severely inhibited the adsorption of CIP by the adsorbent, and the inhibition became stronger with increasing Ca²⁺ concentration. This was due to the fact that competitive adsorption took place between the CIP and limited active sites on the surface of the biochar via electrostatic interaction (Che et al., 2023). High concentrations of Cd²⁺ promoted CIP adsorption, while low concentrations inhibited CIP adsorption for the RBC, and the opposite occurred for the Mn-RBC. The minimal impacts of the four cations on the CIP adsorption by the Mn/Na-RBC indicate its excellent resistance to interference and its selective adsorption capacity in natural water bodies, making it a promising adsorbent material.

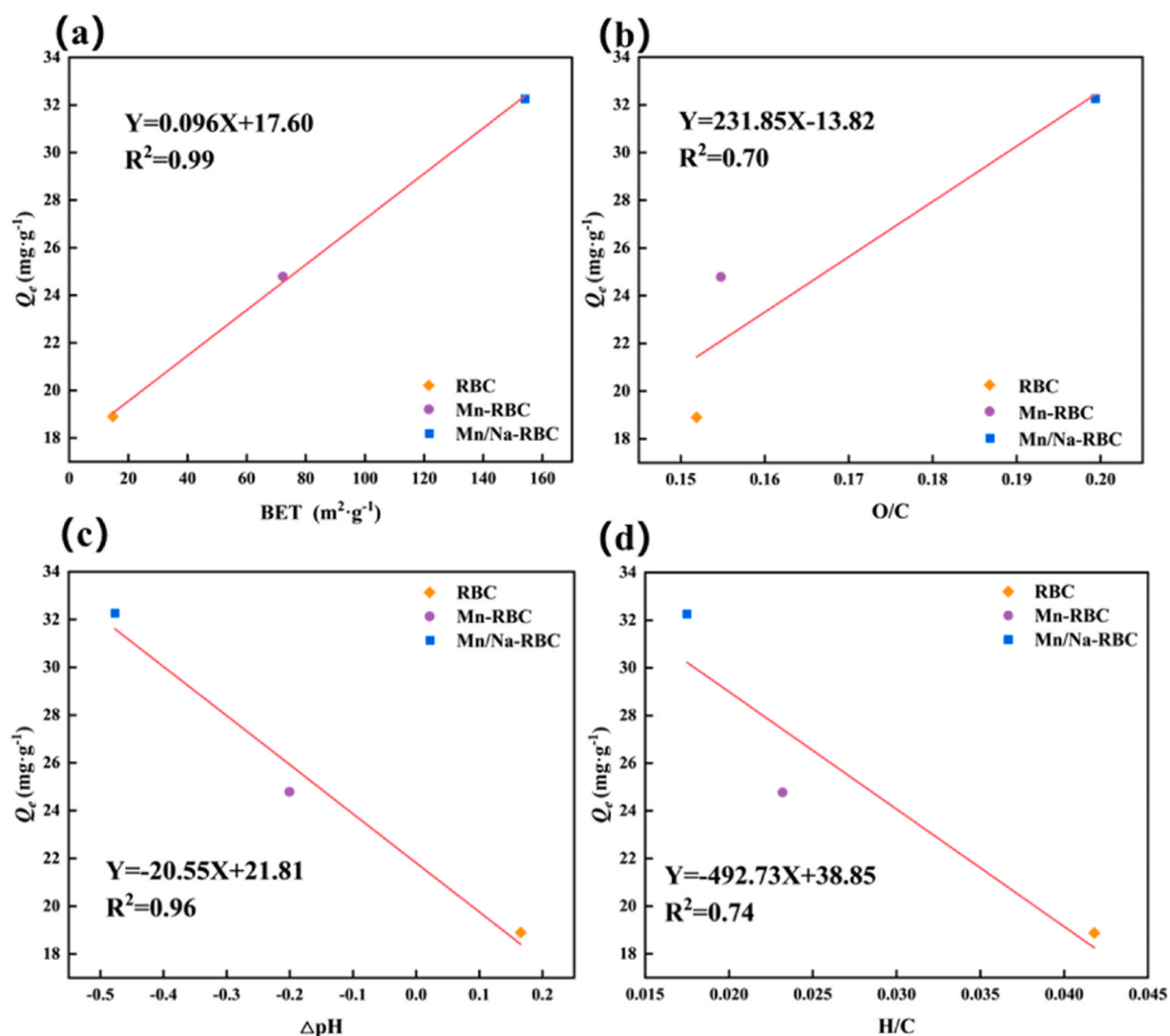


Fig. 7. Correlations between the maximum adsorption capacity ($q_{\max}$) and the properties of the biochars. BET (a), O/C (b), ΔpH (c) and H/C (d).

3.4. CIP adsorption mechanisms of Mn/Na-RBC

The connection between the physical properties and adsorption capacity could help clarify the behavior and mechanism of the CIP adsorption by Mn/Na-RBC (Fig. 7a–d). For physical adsorption, pore filling could potentially serve as the primary driving force for the adsorption of CIP onto Mn/Na-RBC. This could be due to the substantial formation of micro and meso-pores resulting from the oxidative action of KMnO_4 and the alkaline treatment with NaOH . Furthermore, electrostatic attraction was another significant factor promoting the adsorption of CIP. The alkali modification of the surface introduced functional groups such as $-\text{OH}$ and $-\text{COOH}$. Adsorption occurred between the negatively charged functional groups on the adsorbents and the CIP due to electrostatic attraction. The XRD and element analyses revealed that the emergence of a graphite structure resulted from the high carbonization of the Mn/Na-RBC, enabling it to act as a π -electron donor. The N-heteroaromatic ring of the CIP molecules (π -electron acceptors) could establish π - π conjugation with the biochar. By comparing the FT-IR spectra before and after CIP adsorption (Fig. S2), it was found that the peak of the Mn/Na-RBC shifted from the original 3431–3439 cm^{-1} after CIP adsorption; the peak at 1097 cm^{-1} was also enhanced. These results suggest that the CIP adsorption capacity was enhanced due to hydrogen bonding interactions between the oxygen-containing functional groups ($-\text{OH}$, $\text{C}-\text{O}$) in the Mn/Na-RBC and the CIP molecules. Furthermore, the carbonyl oxygen within the biochar could establish n - π interactions with the aromatic ring of the CIP. This is supported by the FT-IR peak displacement and the width of the $\text{C}=\text{O}$ peak.

According to the XPS spectra after adsorption (Fig. S3), there was a noticeable change in the Mn $2p_{3/2}$ peak, i.e., the centroid shifted toward a higher binding energy. The proportions of Mn(VI) and Mn(III) on the surface increased to 8.60% and 13.35%, respectively, while the proportion of Mn(II) decreased to 53.10%. These changes suggested that a portion of the Mn(II) underwent conversion to Mn

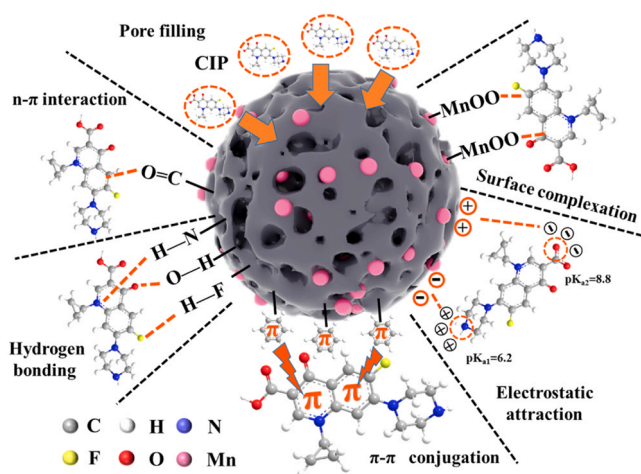


Fig. 8. Schematic diagram of the CIP adsorption mechanism on the Mn/Na-RBC.

(VI) and Mn(III) during the adsorption and removal processes. Under acidic conditions, the MnO_2 was present as a complex acid on the surface of the biochar and underwent hydrolysis to produce hydrated MnOOH minerals (Zhang et al., 2021). These hydrated minerals formed biochar- MnOOH complexes on the surface of the Mn/Na-RBC, implying that Mn-mediated surface complexation was likely the primary factor driving the chemisorption.

Based on the above analysis, Mn/Na-RBC modification greatly enhanced the CIP adsorption capacity through pore filling and electrostatic interactions, followed by π - π conjugation, n - π interactions, H-bonding, and surface complexation mechanisms. The potential adsorption mechanisms are shown in Fig. 8.

4. Conclusions

In this study, impregnation of rice straw followed by pyrolysis (Mn/Na-RBC) led to CIP excellent adsorption. The outstanding adsorption capacity of the Mn/Na-RBC, reaching a maximum of $32.25 \text{ mg} \cdot \text{g}^{-1}$, was primarily due to its substantial specific surface area ($154.12 \text{ m}^2 \cdot \text{g}^{-1}$) and a greater number of effective adsorption sites. In contrast to the RBC, the Mn/Na-RBC had a higher aromaticity, polarity and more abundant oxygen-containing functional groups. The isotherms, thermodynamics, and kinetics data revealed that the adsorption of CIP onto the Mn/Na-RBC was a spontaneous and exothermic multilayer physicochemical adsorption process. The results of the adsorption experiments and characterization analyses provide evidence that the primary adsorption mechanisms include pore filling, electrostatic interaction, π - π conjugation, n - π interactions, H-bonding, and surface complexation. The results of this study indicate that Mn/Na-RBC may be a promising adsorbent for effectively removing ciprofloxacin from aqueous solutions.

CRedit authorship contribution statement

Runjuan Zhou: Supervision. Yingke Sun: Data curation. Penghui Li: Software. Ming Zhang: Supervision. Zhanao Lv: Writing – review & editing. Shuai Shao: Writing – original draft, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.eti.2024.103611](https://doi.org/10.1016/j.eti.2024.103611).

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